

CGM Glucose Forecasting on OhioT1DM: Combined Findings Report

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Dataset: OhioT1DM, all 12 patients (6 from the 2018 release: 559, 563, 570, 575, 588, 591; 6 from the 2020 release: 540, 544, 552, 567, 584, 596) **Horizons:** 15 / 30 / 60 minutes ahead **Lookback window:** 60 minutes (12 steps at 5-minute sampling) **Repository:** CGM-Forecasting-Research

This report combines two separate experiments run on the same 12-patient cohort: (1) a **feature-set ablation** (does adding heart rate, meals, sleep, or insulin data to a fixed LSTM improve forecasts?), and (2) an **architecture comparison** (does a different model — GRU, BiLSTM, TCN, Transformer — outperform a plain LSTM on the best available feature set?). Both patient-wise results and mean $\pm$ SD summaries are included, per the original brief, so an average never hides a patient-level outlier without being flagged.

1. Executive Summary

- **Auxiliary signals (HR, carbs, sleep, insulin) give at most a small, mostly non-significant boost** over glucose-only forecasting at 15–30 min horizons. The one clear exception: **meal/carb data significantly improves 60-min forecasts** ($p=0.021$).
 - **Architecture matters more than extra features did.** Swapping the model architecture (holding a fixed feature set) produced larger, statistically robust differences than swapping feature sets did.
 - **GRU is the best-performing and safest architecture at every horizon tested**, significantly beating LSTM, BiLSTM, TCN, and Transformer at 15/30 min, and beating BiLSTM/TCN/Transformer (but tying LSTM) at 60 min.
 - **TCN and Transformer underperform simple recurrent models across the board** — a clean, consistent, and clinically meaningful finding given the limited lookback window (60 min) and cohort size (12 patients).
 - **Two patients (540, 567 — both 2020 cohort) are consistently the hardest to forecast** across every architecture and every horizon, and patient 540 also has the worst clinical-safety numbers. This should not be averaged away.
 - **2018 vs. 2020 cohort differences are not statistically significant** (Mann-Whitney $p=0.39$ – 0.82 across horizons) for glucose-only forecasting, supporting pooling them — but 2020 patients cannot use HR/Steps features (different wristband; no data), a hard dataset limitation, not a modeling choice.
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2. Experiment 1 — Feature-Set Ablation (fixed LSTM architecture)

Protocol: a single LSTM (64 units) trained separately per patient, per horizon, per feature set. Feature sets B, D, F depend on HeartRate/Steps, which are 100% missing for all 6 2020 patients (different wristband hardware — confirmed by directly inspecting the XML, zero basis_heart_rate/basis_steps events) — so those three feature sets ran on the 6 2018 patients only. Feature sets A, C, E ran on all 12.

Feature Set	Channels
A	Glucose only
B	Glucose + HeartRate (2018 only, n=6)
C	Glucose + Carbs (meals)
D	Glucose + HR + Steps + Carbs + Sleep (2018 only, n=6)
E	Glucose + Insulin (bolus + basal)
F	Full multimodal: HR + Steps + Carbs + Sleep + Insulin (2018 only, n=6)

2.1 Summary (mean $\pm$ SD across patients)

Horizon	Model	N	RMSE	MAE	R2
15min	A: Glucose only	12	16.03 $\pm$ 2.56	10.99 $\pm$ 1.15	0.92 $\pm$ 0.03
15min	B: Glucose + HeartRate	6	15.70 $\pm$ 2.11	10.48 $\pm$ 1.16	0.92 $\pm$ 0.03
15min	C: Glucose + Carbs(meals)	12	15.83 $\pm$ 2.74	10.78 $\pm$ 1.24	0.92 $\pm$ 0.03
15min	D: Glucose + HR + Steps + Carbs + Sleep	6	15.45 $\pm$ 2.32	10.45 $\pm$ 0.99	0.92 $\pm$ 0.03
15min	E: Glucose + Insulin (bolus+basal)	12	15.97 $\pm$ 2.63	10.80 $\pm$ 1.28	0.92 $\pm$ 0.03
15min	F: Full multimodal (HR+Steps+Carbs+Sleep+Insulin)	6	15.99 $\pm$ 2.31	11.07 $\pm$ 1.20	0.92 $\pm$ 0.02
30min	A: Glucose only	12	24.35 $\pm$ 3.66	17.31 $\pm$ 2.34	0.81 $\pm$ 0.05
30min	B: Glucose + HeartRate	6	23.45 $\pm$ 3.10	16.67 $\pm$ 2.10	0.82 $\pm$ 0.06
30min	C: Glucose + Carbs(meals)	12	23.77 $\pm$ 3.61	16.92 $\pm$ 2.19	0.82 $\pm$ 0.05
30min	D: Glucose + HR + Steps + Carbs + Sleep	6	22.92 $\pm$ 2.72	16.32 $\pm$ 1.42	0.83 $\pm$ 0.05
30min	E: Glucose + Insulin (bolus+basal)	12	23.95 $\pm$	17.18 $\pm$	0.81 $\pm$

Horizon	Model	N	RMSE	MAE	R2
			3.15	1.82	0.05
30min	F: Full multimodal (HR+Steps+Carbs+Sleep+Insulin)	6	23.04 ± 2.49	16.51 ± 1.52	0.82 ± 0.06
60min	A: Glucose only	12	36.89 ± 4.95	27.75 ± 3.73	0.56 ± 0.11
60min	B: Glucose + HeartRate	6	35.77 ± 4.13	26.54 ± 2.79	0.58 ± 0.12
60min	C: Glucose + Carbs(meals)	12	36.14 ± 5.34	27.08 ± 4.07	0.58 ± 0.11
60min	D: Glucose + HR + Steps + Carbs + Sleep	6	34.92 ± 3.43	25.86 ± 2.68	0.60 ± 0.10
60min	E: Glucose + Insulin (bolus+basal)	12	36.15 ± 4.78	27.03 ± 3.54	0.58 ± 0.10
60min	F: Full multimodal (HR+Steps+Carbs+Sleep+Insulin)	6	36.66 ± 2.72	27.03 ± 2.14	0.55 ± 0.13

2.2 Statistical significance (Wilcoxon signed-rank, paired by patient)

Across all 12 patients (A vs. C vs. E):

Horizon	Comparison	p-value	Result
15min	A vs C	0.24	not significant
15min	A vs E	0.63	not significant
30min	A vs C	0.11	not significant
30min	A vs E	0.85	not significant
60min	A vs C	0.021	C significantly better (carbs help at 60min)
60min	A vs E	0.70	not significant

Within the 6 2018 patients only (does HR/Steps/full multimodal beat glucose-only?):

Horizon	A vs B (HeartRate)	A vs D (full wearable)	A vs F (+ insulin)
15min	p=0.44 (B better 4/6)	p=0.16 (D better 5/6)	p=0.84 (F better 3/6)
30min	p=0.69 (B better 4/6)	p=0.56 (D better 4/6)	p=0.44 (F better 4/6)
60min	p=0.69 (B better 4/6)	p=0.16 (D better 5/6)	p=0.44 (F better 2/6)

Interpretation: none of these 6-patient comparisons reach significance — likely underpowered at n=6 rather than a true null result, given D wins in 5/6 patients at two horizons. “D: full wearable + meals + sleep” is the most promising direction worth revisiting

with more patients (once 2020-cohort-comparable wearable data or a larger cohort is available), but should not yet be claimed as proven.

3. Experiment 2 — Architecture Comparison (fixed feature set)

Protocol: LSTM, GRU, BiLSTM, TCN, and Transformer, each trained separately per patient, per horizon, on the same feature set (Glucose + carbs + sleep + insulin — the subset available for all 12 patients). All 12 patients included in every architecture/horizon cell (n=12 throughout, unlike Experiment 1).

3.1 Summary (mean $\pm$ SD across all 12 patients)

Horizon	Architecture	N	RMSE	MAE	MARD	TimeLag_min	CEGA_A_%	CEGA_B_%	CEGA_C_%	CEGA_D_%	CEGA_E_%
15min	BiLSTM	1	18.71	13.12	8.94 $\pm$	0.00 $\pm$	90.05	8.50 $\pm$	0.05	1.41 $\pm$	0.00
		2	$\pm$ 3.16	$\pm$ 2.09	1.62	0.00	$\pm$ 3.99	3.50	$\pm$ 0.09	1.12 $\pm$	0.00
15min	GRU	1	17.32	12.00	8.10 $\pm$	0.00 $\pm$	92.33	6.52 $\pm$	0.04	1.11 $\pm$	0.00
		2	$\pm$ 2.75	$\pm$ 1.53	1.22	0.00	$\pm$ 2.73	2.38	$\pm$ 0.09	0.89 $\pm$	0.00
15min	LSTM	1	18.81	13.21	9.01 $\pm$	0.00 $\pm$	90.00	8.50 $\pm$	0.04	1.46 $\pm$	0.00
		2	$\pm$ 3.40	$\pm$ 2.51	1.73	0.00	$\pm$ 3.94	3.12	$\pm$ 0.09	1.33 $\pm$	0.00
15min	TCN	1	24.89	18.15	12.33	0.00 $\pm$	81.99	15.52	0.05	2.44	0.00
		2	$\pm$ 5.85	$\pm$ 4.52	$\pm$ 3.13	0.00	$\pm$ 7.94	$\pm$ 6.93	$\pm$ 0.14	$\pm$ 1.87	$\pm$ 0.00
15min	Transformer	1	31.04	22.85	15.86	0.00 $\pm$	72.87	24.11	0.18 $\pm$	2.83	0.01
		2	$\pm$ 4.66	$\pm$ 3.20	$\pm$ 3.01	0.00	$\pm$ 8.01	$\pm$ 6.57	0.33	$\pm$ 1.97	$\pm$ 0.05
30min	BiLSTM	1	26.16	19.03	13.16	0.00 $\pm$	79.57	17.68	0.15 $\pm$	2.60	0.00
		2	$\pm$ 3.90	$\pm$ 2.71	$\pm$ 2.71	0.00	$\pm$ 7.36	$\pm$ 5.87	0.22	$\pm$ 1.90	$\pm$ 0.00
30min	GRU	1	24.89	17.75	12.08	0.00 $\pm$	81.95	15.71	0.08	2.26	0.00
		2	$\pm$ 3.35	$\pm$ 2.18	$\pm$ 2.23	0.00	$\pm$ 6.26	$\pm$ 4.84	$\pm$ 0.17	$\pm$ 1.73	$\pm$ 0.00
30min	LSTM	1	25.96	18.80	12.84	0.00 $\pm$	80.35	17.09	0.10	2.46	0.00
		2	$\pm$ 3.85	$\pm$ 2.73	$\pm$ 2.49	0.00	$\pm$ 7.01	$\pm$ 5.66	$\pm$ 0.18	$\pm$ 1.91	$\pm$ 0.00
30min	TCN	1	31.48	23.02	15.64	0.00 $\pm$	73.27	23.58	0.19 $\pm$	2.96	0.00
		2	$\pm$ 4.20	$\pm$ 3.27	$\pm$ 2.90	0.00	$\pm$ 6.72	$\pm$ 5.50	0.23	$\pm$ 1.84	$\pm$ 0.00
30min	Transformer	1	36.58	27.33	19.28	0.00 $\pm$	65.82	30.13	0.30	3.72	0.03
		2	$\pm$ 5.17	$\pm$ 3.80	$\pm$ 4.30	0.00	$\pm$ 9.33	$\pm$ 7.46	$\pm$	$\pm$	$\pm$

Horizon	Architecture	N	RMSE	MAE	MARD	TimeLag_min	CEGA_A_%	CEGA_B_%	CEGA_C_%	CEGA_D_%	CEGA_E_%
									0.36	2.44	0.07
60min	BiLSTM	1	38.01	28.46	19.88	0.00 ±	63.98	31.67	0.31 ±	4.02	0.02
		2	± 5.21	± 3.95	± 3.94	0.00	± 8.91	± 7.11	0.33	±	±
										2.22	0.05
60min	GRU	1	36.78	27.63	19.31	0.00 ±	65.09	30.58	0.28	4.02	0.02
		2	± 4.50	± 3.51	± 3.83	0.00	± 8.73	± 6.39	±	±	±
									0.37	2.89	0.05
60min	LSTM	1	37.40	28.01	19.74	0.00 ±	63.73	31.68	0.30	4.27	0.02
		2	± 5.04	± 3.83	± 4.15	0.00	± 9.74	± 7.55	±	±	±
									0.32	2.62	0.05
60min	TCN	1	41.18	31.24	21.83	0.00 ±	59.94	34.92	0.31 ±	4.79	0.05
		2	± 4.32	± 3.58	± 4.72	0.00	± 8.95	± 6.92	0.28	±	±
										2.36	0.10
60min	Transformer	1	44.38	34.03	24.31	0.00 ±	55.94	38.24	0.38	5.39	0.05
		2	± 6.18	± 5.07	± 4.88	0.00	± 8.60	± 5.92	±	±	± 0.12
									0.44	3.45	

(TimeLag_min is 0.00 ± 0.00 throughout — none of the five architectures showed a detectable prediction lag at this sampling resolution.)

3.2 Statistical significance (Wilcoxon signed-rank, GRU vs. each other architecture, n=12)

Horizon	GRU vs LSTM	GRU vs BiLSTM	GRU vs TCN	GRU vs Transformer
15min	p=0.0015 ** (GRU wins 11/12)	p=0.0015 ** (11/12)	p=0.0005 *** (12/12)	p=0.0005 *** (12/12)
30min	p=0.0210 * (10/12)	p=0.0015 ** (11/12)	p=0.0005 *** (12/12)	p=0.0005 *** (12/12)
60min	p=0.38 (n.s., 7/12)	p=0.0161 * (10/12)	p=0.0005 *** (12/12)	p=0.0005 *** (12/12)

(* p<0.05, ** p<0.01, *** p<0.001)

Interpretation: GRU's advantage over TCN and Transformer is large, consistent, and highly significant at every horizon — it wins every single patient comparison (12/12) against both at all three horizons. GRU's edge over LSTM and BiLSTM is real but smaller, and **disappears entirely between GRU and LSTM at 60 min** (p=0.38) — at longer horizons, the two are statistically indistinguishable.

3.3 Clinical safety — Clarke Error Grid, dangerous zones (D + E)

Horizon	GRU	LSTM	BiLSTM	TCN	Transformer
15min	1.11%	1.46%	1.41%	2.44%	2.84%
30min	2.26%	2.46%	2.60%	2.96%	3.75%
60min	4.04%	4.29%	4.05%	4.84%	5.44% (worst)

GRU has the lowest dangerous-zone rate at every horizon (tied with BiLSTM at 60min). Transformer is worst at every horizon, and its worst single case — **patient 540 at 60 min, 11.8% of predictions in dangerous zones** — is a genuine clinical concern, not a rounding artifact.

4. Per-Patient Consistency (GRU, the recommended architecture)

Averages can hide patient-level failure. Full per-patient GRU results, all three horizons:

Patient	Cohort	RMSE_15min	RMSE_30min	RMSE_60min	MAE_15min	MAE_30min	MAE_60min	MARD_15min	MARD_30min	MARD_60min
559	2018	18.24	24.36	35.63	12.61	16.65	25.8	8.28	11.1	17.52
563	2018	15.47	22.39	33.52	10.75	15.91	24.32	6.73	9.87	15.49
570	2018	16.17	21.93	34.5	11.88	16.09	26.58	5.8	8.07	13.16
575	2018	19.79	27.26	38.3	12.2	17.87	27.62	8.35	12.2	20.04
588	2018	20.32	26.33	36.28	14.85	19.35	27.21	9.11	11.73	16.41
591	2018	17.6	25.68	37.15	12.21	18.78	28.91	9.49	14.36	23.71
540	2020	16.97	26.52	44.37	12.65	20.19	33.59	8.96	14.54	23.06
544	2020	13.99	20.75	32.68	9.82	14.75	24.39	6.55	9.66	16.62
552	2020	13.26	22.8	32.14	10.01	16.81	24.5	7.8	12.8	19.03
567	2020	20.56	30.9	42.87	13.29	21.58	32.59	9.66	15.81	25.71
584	2020	21.21	29.34	42.88	13.39	19.8	32.37	8.77	13.11	22.94

Patient	Cohort	RMSE_15min	RMSE_30min	RMSE_60min	MAE_15min	MAE_30min	MAE_60min	MARD_15min	MARD_30min	MARD_60min
596	2020	14.29	20.41	31.06	10.33	15.19	23.7	7.7	11.74	18.05

Range at 60 min: 31.1 (patient 596) to 44.4 mg/dL (patient 540) — a 13.3 mg/dL spread across otherwise-comparable patients using the identical pipeline.

Consistently hardest patients across every architecture and horizon: 540 and 567 (both 2020 cohort). Patient 540 combines the worst RMSE *and* the worst clinical-safety numbers at 60 min — worth a closer look (unusual glycemic variability, sensor data quality, or a genuinely harder physiological case) rather than treating it as noise the mean average washes out. Patient 591 (2018 cohort) is a consistent secondary hard case, particularly at 60 min.

No systematic 2018-vs-2020 cohort gap was found in glucose-only forecasting (Mann-Whitney $p=0.39-0.82$ across horizons) — the difficulty is patient-specific, not cohort-specific.

5. Combined Interpretation

Putting both experiments together: **the choice of architecture had a bigger, more statistically robust effect on forecast accuracy than the choice of input features did**, within the ranges tested here. This is a genuinely useful, somewhat counterintuitive finding — it suggests that before investing further effort in acquiring or engineering more physiological input channels (more sensors, more derived features), there's more headroom in refining model design (this GRU result, further architecture search, or hyperparameter tuning of GRU itself) and in expanding the patient cohort, particularly to shore up the 6-patient feature-set comparisons that are currently underpowered.

The two-patient “hard case” cluster (540, 567) is worth a targeted follow-up — patient-wise clinical review or per-patient hyperparameter adjustment — rather than folding into an already-acceptable-looking mean.

6. Known Limitations

- **Feature-set comparisons for B/D/F run on only 6 patients** (2018 cohort) — likely underpowered (see §2.2); the promising “D: full wearable” direction needs a larger sample before being treated as established.
- **CEGA zone classification and MARD/time-lag are only computed in Experiment 2** (architecture comparison) — Experiment 1 (feature-set) only reports RMSE/MAE/ R^2 . A useful follow-up would be re-running the feature-set ablation with the full clinical metric suite for direct comparability.
- **N-BEATS/N-HiTS and TFT were not included** in this architecture comparison — N-BEATS needs a univariate-only variant (doesn't take the multivariate feature set the

same way), and TFT needs a different stack (pytorch-forecasting) than the Keras models used here. Both remain open next steps.

- **Single train/val/test split per patient** (no cross-validation) — results reflect one held-out test period per patient, per the dataset's official split, not a K-fold estimate.
 - **Shanghai dataset cross-validation not yet performed** — this report covers OhioT1DM only; extending the architecture comparison to Shanghai (once its preprocessing pipeline is finalized) is the natural next step for testing generalizability across datasets, as originally scoped.
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Appendix: Data Files

- `featureset_patient_wise.csv` — full per-patient results, Experiment 1 (feature-set ablation)
- `architecture_patient_wise.csv` — full per-patient results, Experiment 2 (architecture comparison)
- `feat_summary.csv` / `arch_summary.csv` — the mean $\pm$ SD summary tables in machine-readable form
- `gru_patient_wise.csv` — the per-patient GRU breakdown used in §4
- `architecture_comparison_chart.png` — mean RMSE by architecture across horizons
- `per_patient_rmse.png` — per-patient RMSE spread, glucose-only baseline